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Vereinfachung mittels Karnaugh-Tabellen

KAR

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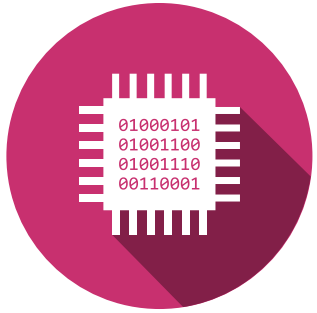
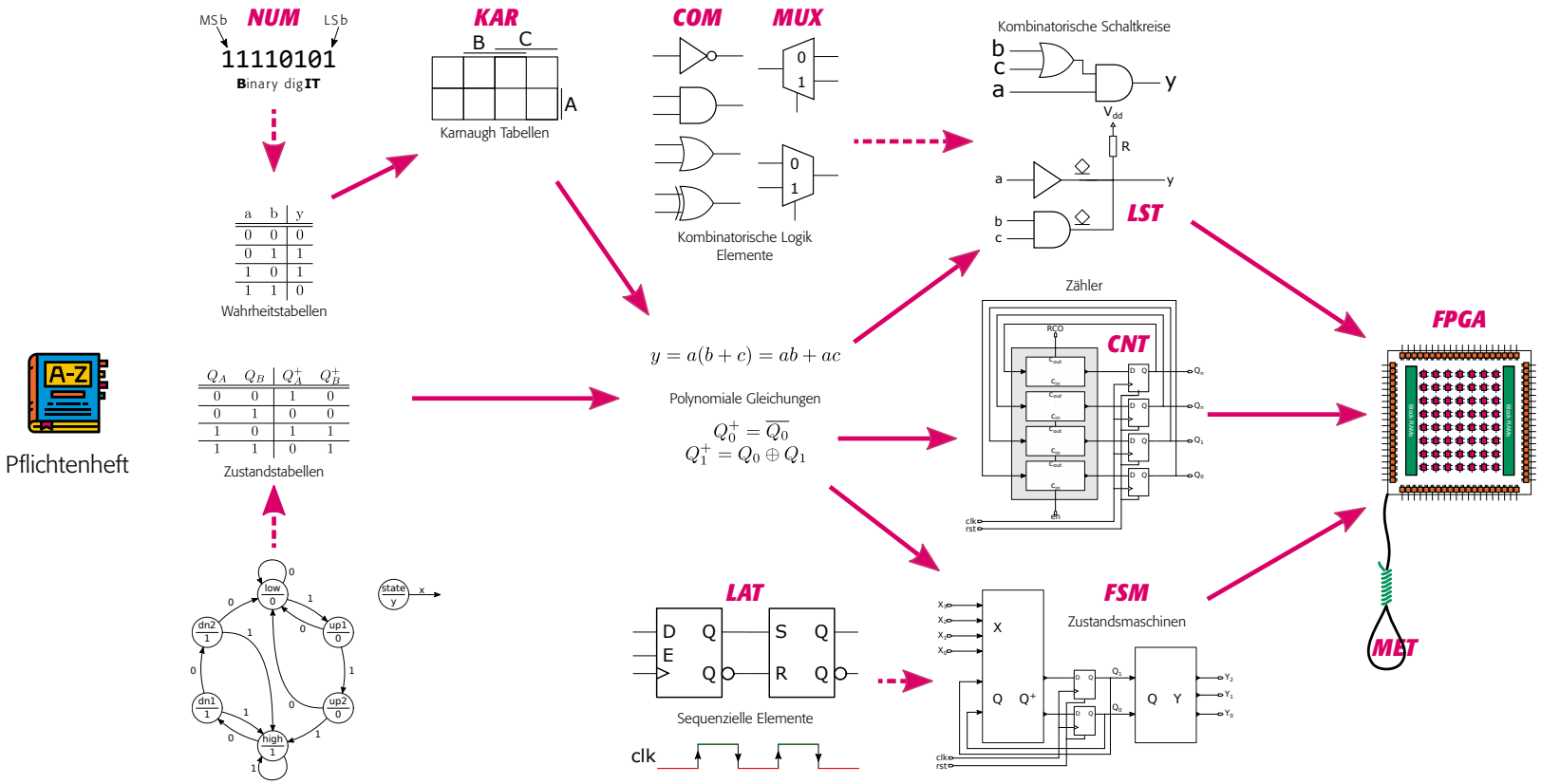


Aktueller Inhalt des Themas im Kurs



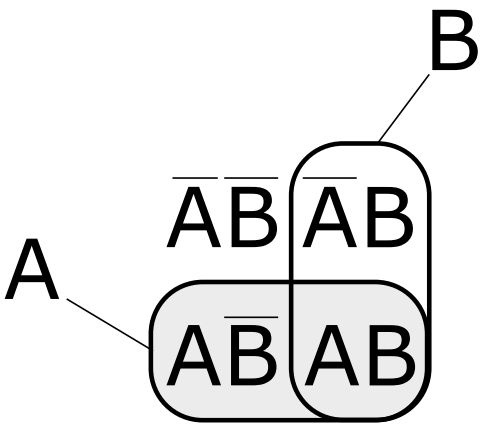
Inhalt

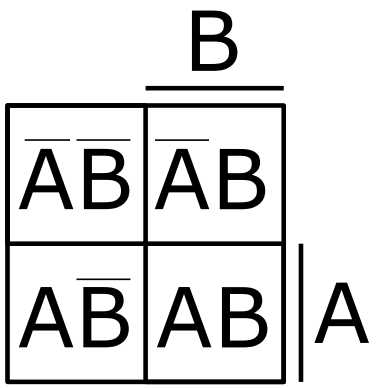
- **Karnaugh Tabellen**
  - mit 2 Variablen
  - mit 3 Variablen
  - mit 4 Variablen
  - Mit mehr als 4 Variablen
- Vereinfachung mittels einer Summe von Produkten
- Vereinfachung mittels Exklusiv-ODER Funktionen
- Funktionen mit einer grossen Anzahl an Eingängen

Karnaugh Tabellen

Mit 2 Variablen

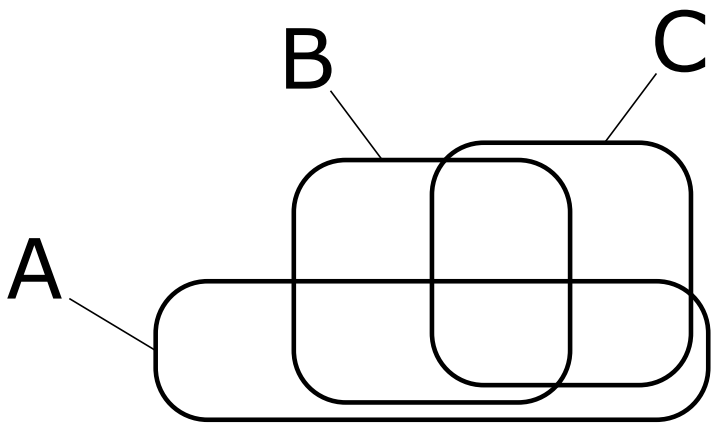
Es handelt sich um ein Venn Diagram

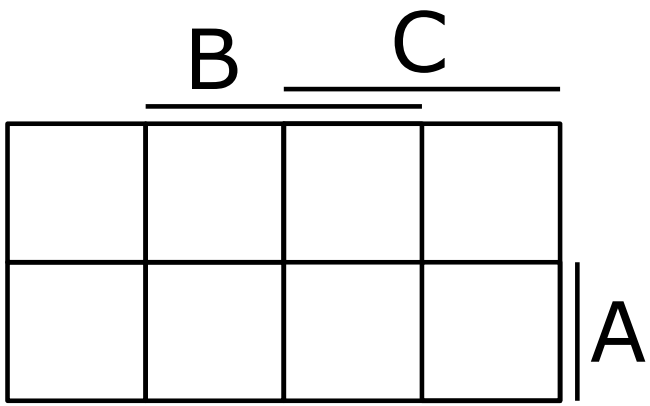




Karnaugh Tabellen

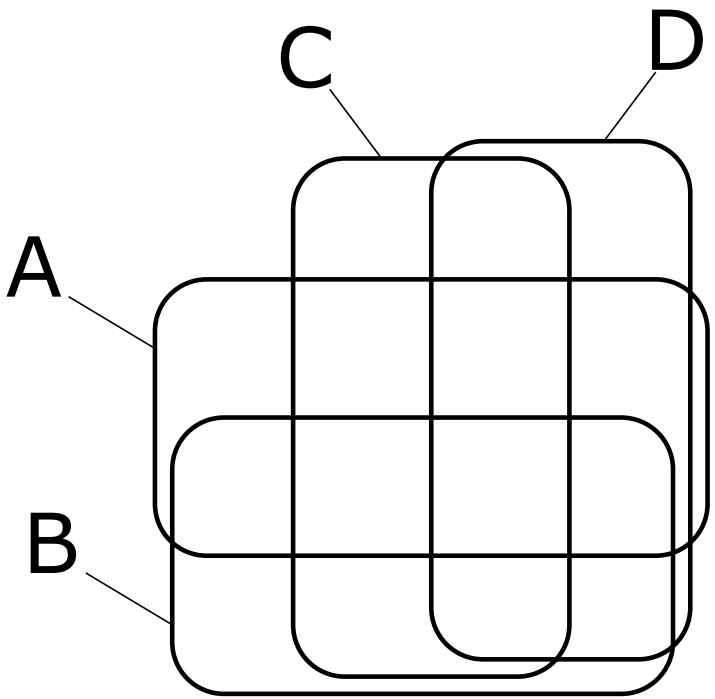
Mit 3 Variablen

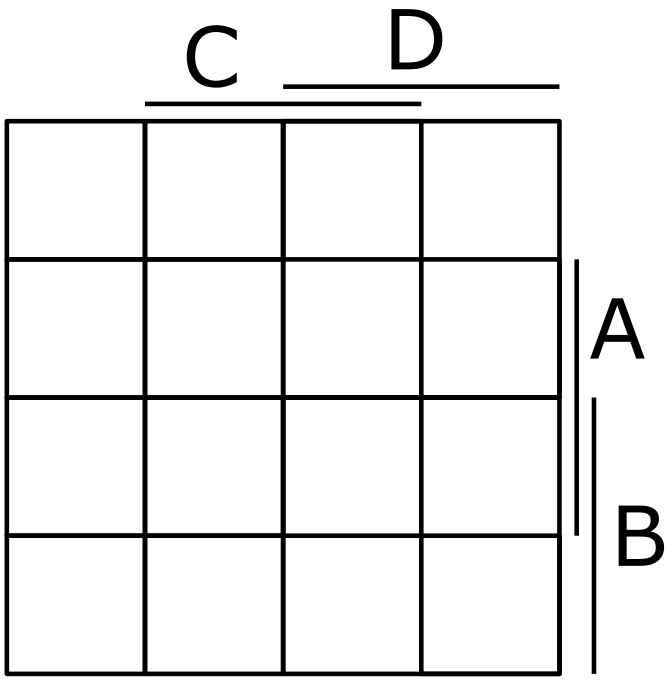




Karnaugh Tabellen

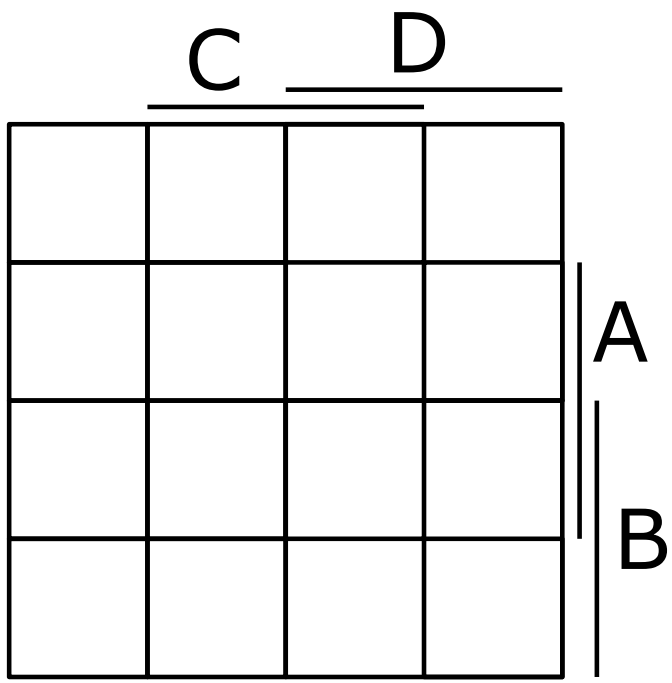
Mit 4 Variablen

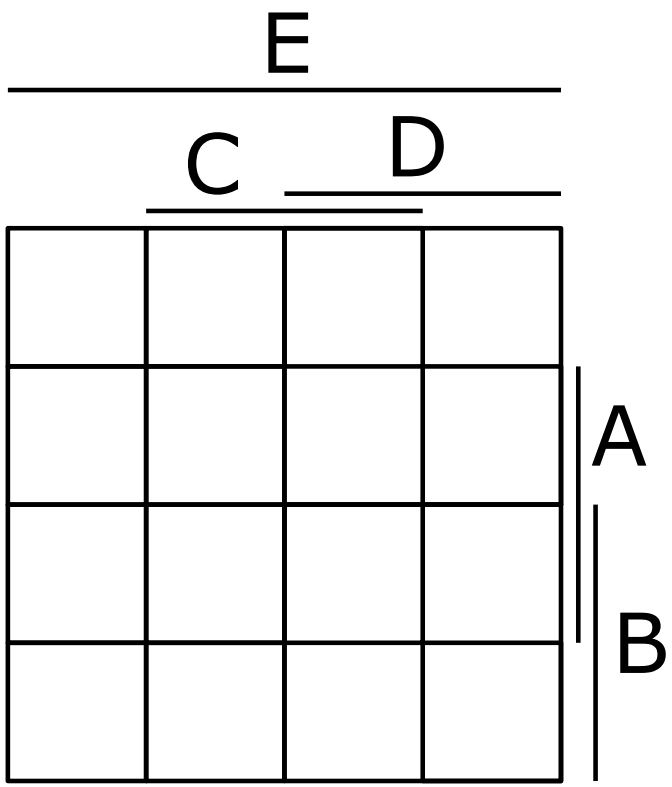




Karnaugh Tabellen

Mit 5 Variablen





Inhalt

- Karnaugh Tabellen
- **Vereinfachung mittels einer Summe von Produkten**
  - Monome
  - Polynome
  - Primimplikant, Wesentlicher Primimplikant
  - Vereinfachungen
  - Unvollständig definierte Funktionen
- Vereinfachung mittels Exklusiv-ODER Funktionen
- Funktionen mit einer grossen Anzahl an Eingängen

Karnaugh Tabellen

Monome

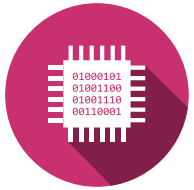
	C		D	
0	0	0	0	
0	0	0	0	A
1	1	1	1	
0	0	0	0	B

BA

	C		D	
0	0	0	0	
1	0	0	1	A
0	0	0	0	
0	0	0	0	B

$\overline{C}\overline{B}A$

Aufgabe 1.1

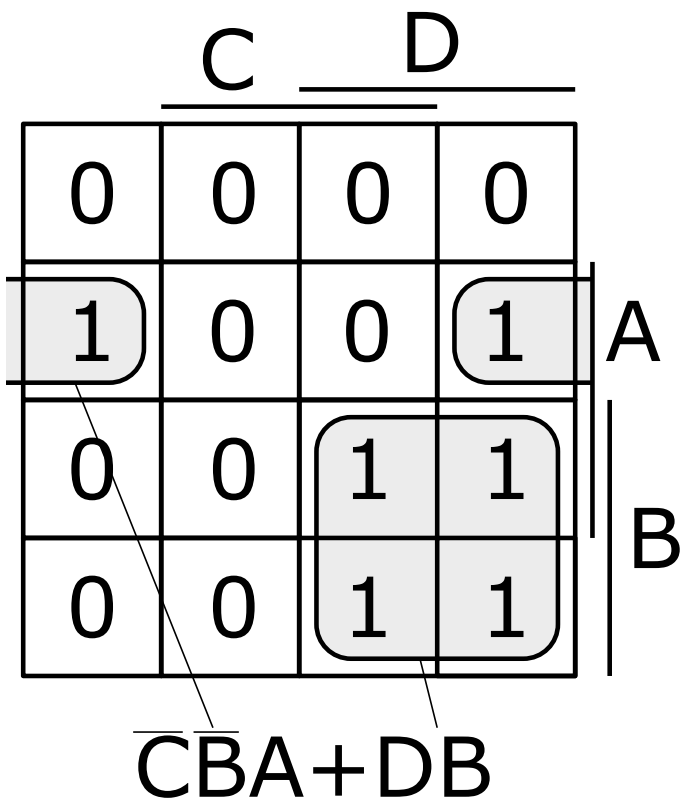


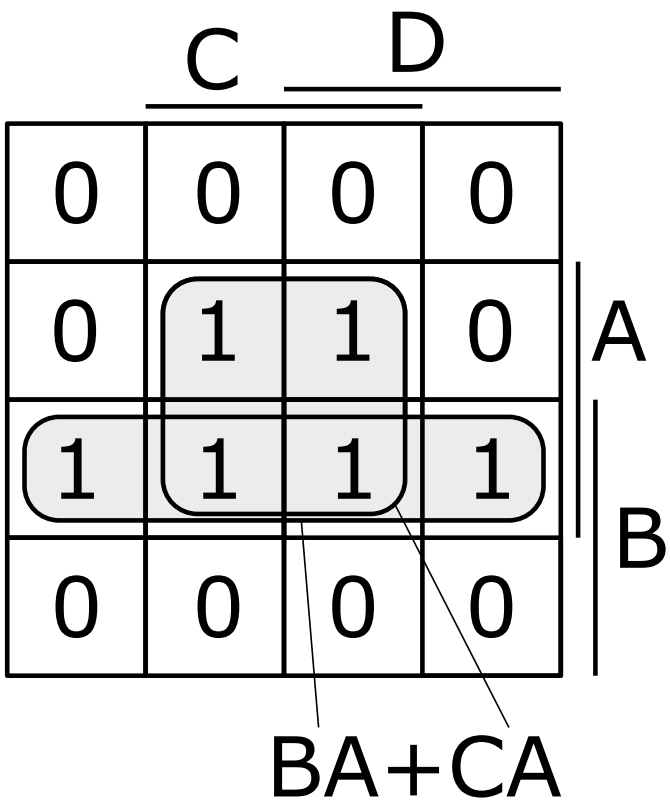
Stellen Sie die folgende Monome in einer Karnaugh-Tafel mit 4 Variablen dar.

$$y_1 = \bar{b}a$$
$$y_2 = \bar{d}\bar{a}$$
$$y_3 = \bar{d}cb$$
$$y_4 = dba$$
$$y_5 = \bar{c}\bar{b}\bar{a}$$
$$y_6 = dcb\bar{a}$$

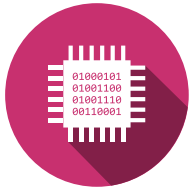
Karnaugh Tabellen

Polynome





Aufgabe 1.3

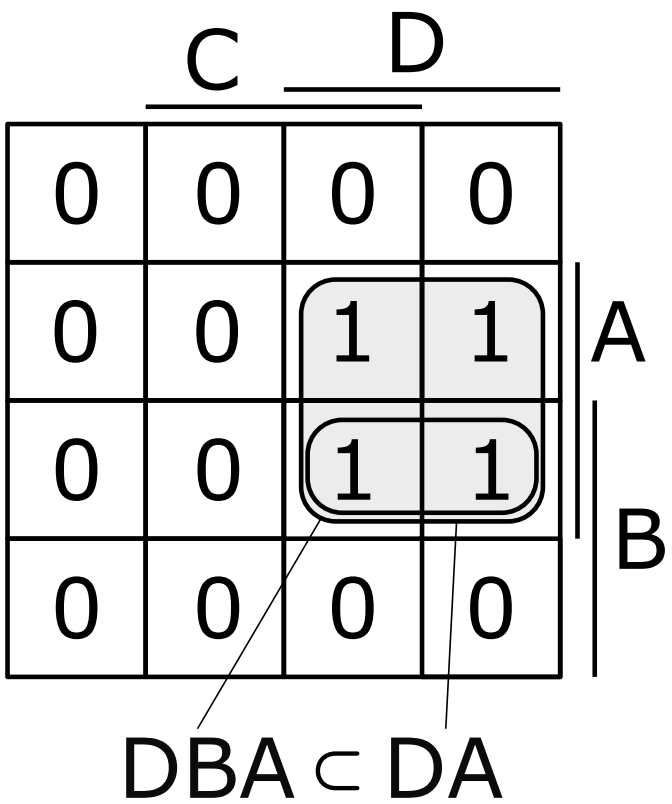


Stellen Sie die folgende Monome in einer Karnaugh-Tafel mit 4 Variablen dar.

$$y_1 = \bar{b} + ac$$
$$y_2 = \bar{d} + \bar{a} + bc$$
$$y_3 = \bar{d}cb + d\bar{c}\bar{b}$$
$$y_4 = db + ab$$
$$y_5 = \bar{c}\bar{b}\bar{a} + \bar{c}\bar{b}$$
$$y_6 = dcb\bar{a} + \bar{d}cb\bar{a}$$

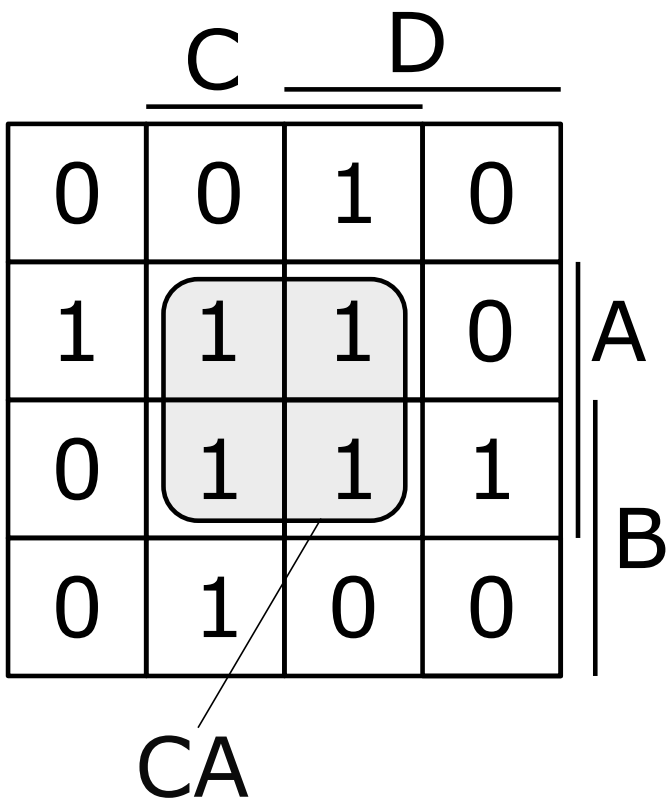
Karnaugh Tabellen

Primimplikant



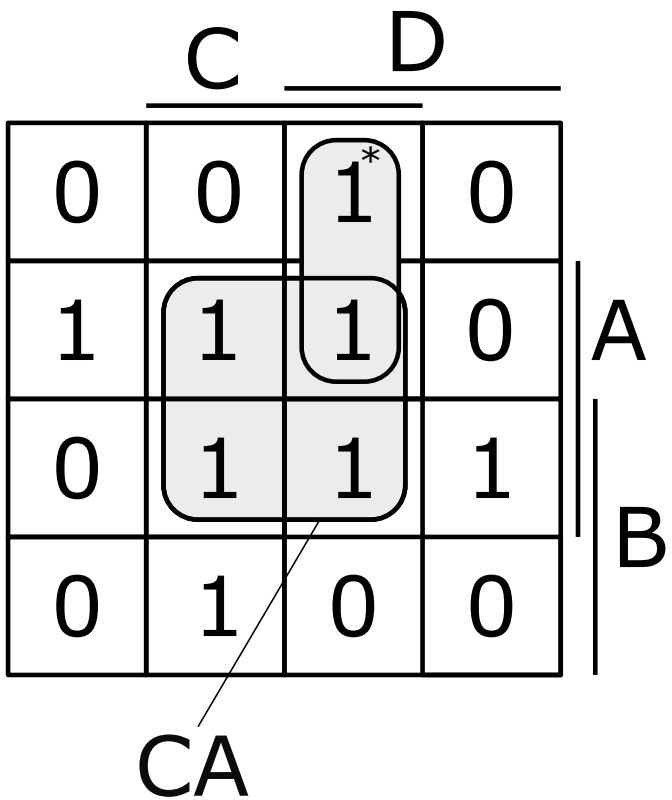
Karnaugh Tabellen

Wesentlicher Primimplikant



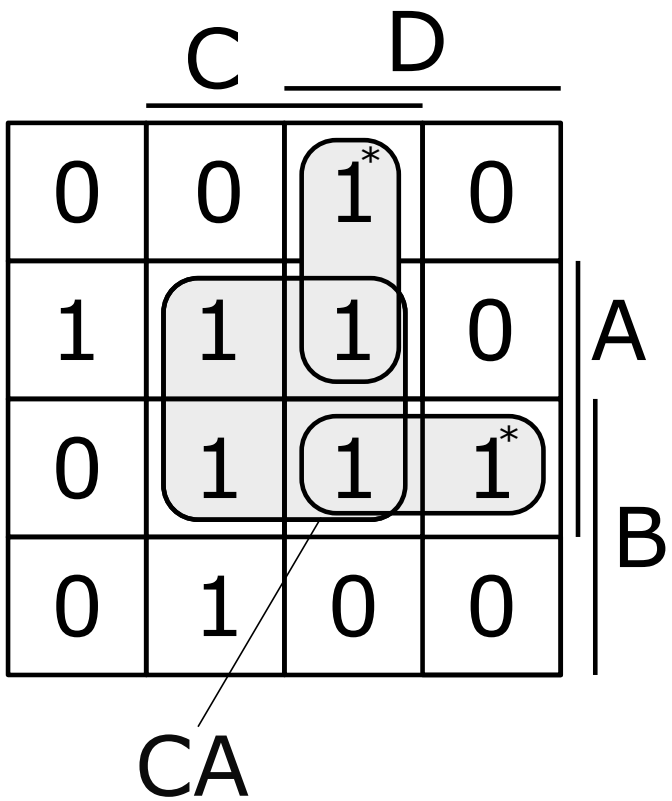
Karnaugh Tabellen

Wesentlicher Primimplikant



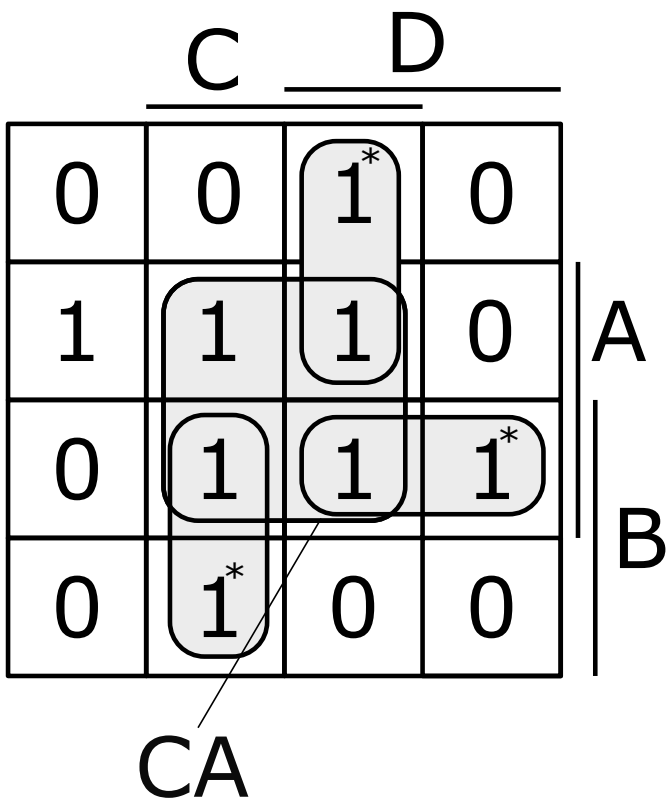
Karnaugh Tabellen

Wesentlicher Primimplikant



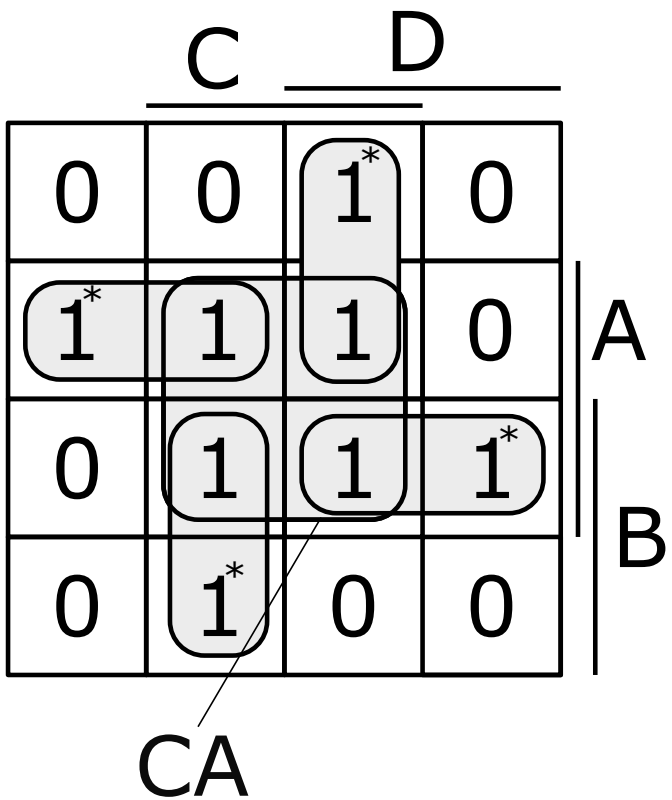
Karnaugh Tabellen

Wesentlicher Primimplikant



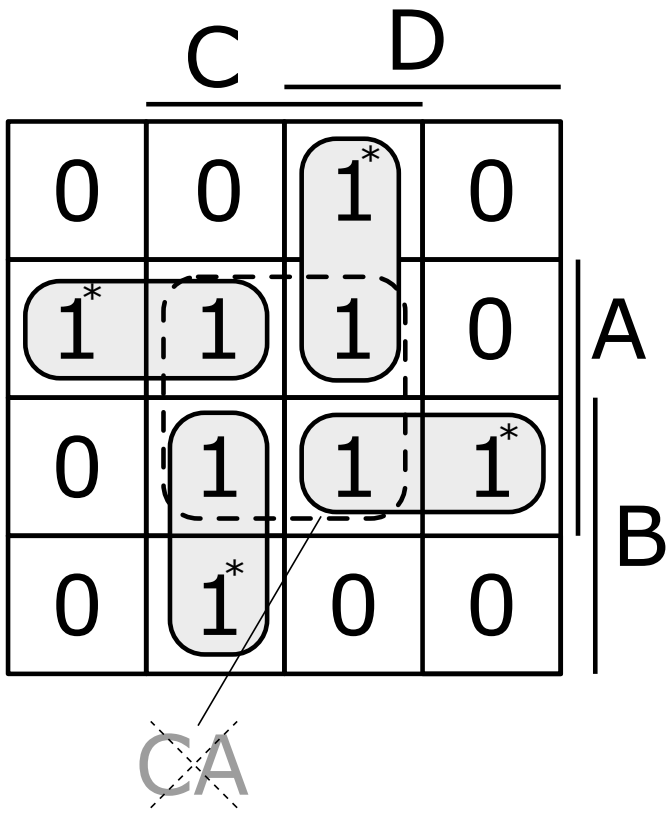
Karnaugh Tabellen

Wesentlicher Primimplikant



Karnaugh Tabellen

Wesentlicher Primimplikant



Karnaugh Tabellen

Bestimmen Sie die minimale Polynomgleichung

	C	D	
0	0	1	0
1	0	1	1
1	0	0	0
0	0	0	1

A

B

Karnaugh Tabellen

Bestimmen Sie die minimale Polynomgleichung

0	0	1*	0
1	0	1	1
1	0	0	0
0	0	0	1

C D A B

$DC\overline{B}^{*}$

→Suche nach wesentlichen Primimplikanten

Karnaugh Tabellen

Bestimmen Sie die minimale Polynomgleichung

	$\overline{C}\,\overline{D}$	$\overline{C}D$	CD	$C\overline{D}$
$\overline{A}\,\overline{B}$	0	0	1*	0
$\overline{A}B$	1	0	1	1
$A\overline{B}$	1*	0	0	0
AB	0	0	0	1

$$DC\overline{B}^{*} + \overline{D}\,\overline{C}A^{*}$$

→Suche nach wesentlichen Primimplikanten

Karnaugh Tabellen

Bestimmen Sie die minimale Polynomgleichung

	$\overline{C}\,\overline{D}$	$\overline{C}D$	$C\overline{D}$	CD
$\overline{A}\,\overline{B}$	0	0	1*	0
$\overline{A}B$	1	0	1	1
$A\overline{B}$	1*	0	0	0
AB	0	0	0	1*

$$D\overline{C}\overline{B}^{*} + \overline{D}\,\overline{C}\overline{A}^{*} + \overline{D}\,\overline{C}B\overline{A}^{*}$$

→Suche nach wesentlichen Primimplikanten

Karnaugh Tabellen

Bestimmen Sie die minimale Polynomgleichung

	$\overline{C}\,\overline{D}$	$\overline{C}D$	$C\overline{D}$	CD
$\overline{A}\,\overline{B}$	0	0	1*	0
$\overline{A}B$	1	0	1	1
$A\overline{B}$	1*	0	0	0
AB	0	0	0	1*

$$DC\overline{B}^{*} + \overline{D}\,\overline{C}A^{*} + D\overline{C}B\overline{A}^{*} + D\overline{B}A$$

→Suche nach den wesentlichen Primimplikanten und danach den verbleibenden Primimplikanten

Karnaugh Tabellen

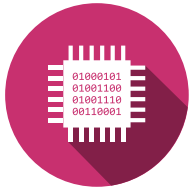
Bestimmen Sie die minimale Polynomgleichung

	$\overline{C}\,\overline{D}$	$\overline{C}D$	CD	$C\overline{D}$
$\overline{A}\,\overline{B}$	0	0	1^*	0
$\overline{A}B$	1	0	1	1
$A\overline{B}$	1^*	0	0	0
AB	0	0	0	1^*

$$DC\overline{B}^* + \overline{D}\,\overline{C}A^* + D\overline{C}B\overline{A}^* + D\overline{B}A + \overline{C}\,\overline{B}A$$

→Suche nach den wesentlichen Primimplikanten und danach den verbleibenden Primimplikanten

Aufgabe 3.1



Bestimmen Sie die minimale Polynomialform der Karnaugh-Tabelle aus 4 Variablen

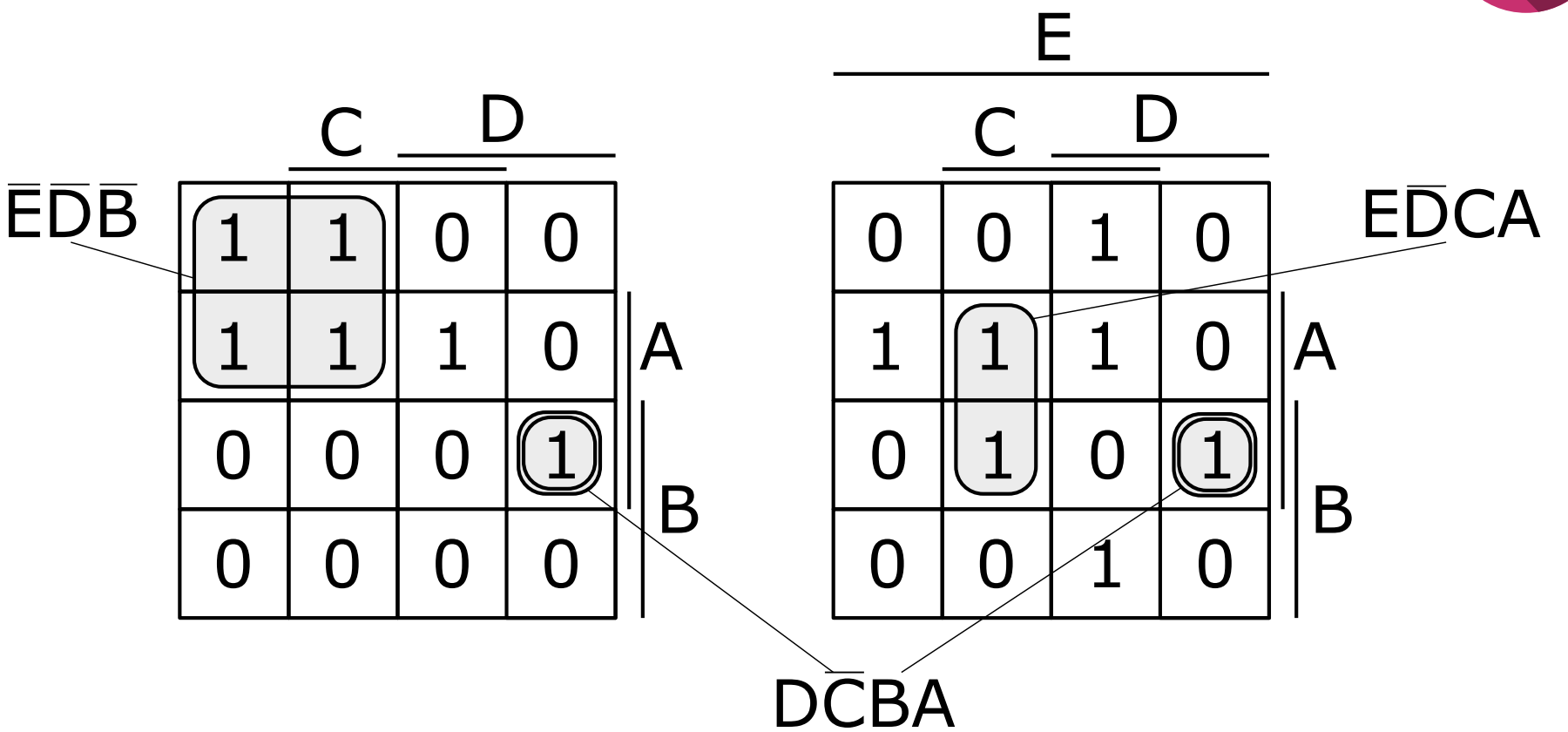
	C		D
1	0	0	1
1	0	1	1
1	0	0	0
1	1	1	1

A

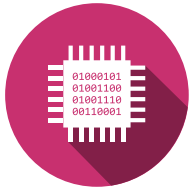
B

Karnaugh Tabellen

Vereinfachung mit 5 Variablen



Aufgabe 3.2



Bestimmen Sie die minimale Polynomialform der Karnaugh-Tabelle aus 5 Variablen

1	1	0	0
1	1	1	0
0	0	0	1
0	0	0	0

E

C D

0	0	1	0
1	1	1	0
0	1	0	1
0	0	1	0

A

B

Karnaugh Tabellen

Unvollständig definierte Funktion

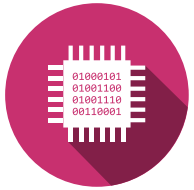
	C		D
1	1	-	1
1	0	-	1
1	1	-	-
1	0	-	-

A

B

$$\overline{C} + \overline{B}\,\overline{A} + BA$$

Aufgabe 3.11



Bestimmen Sie die minimale Polynomialform der Mehrheitsfunktion mit 4 Eingängen:

- die Funktion ergibt eine '0', wenn mehr als die Hälfte der Eingänge auf '0' sind.
- die Funktion ergibt eine '1', wenn mehr als die Hälfte der Eingänge auf '1' sind,
- wenn gleich viele Eingänge auf '0' und auf '1' sind, dann kann das Resultat irgendwelchen Wert nehmen.

Inhalt

- Karnaugh Tabellen
- Vereinfachung mittels einer Summe von Produkten
- **Vereinfachung mittels Exklusiv-ODER Funktionen**
  - Dastellung
  - Vereinfachung
- Funktionen mit einer grossen Anzahl an Eingängen

Karnaugh Tabellen

Darstellung einer Exklusiv-Oder (XOR) Funktion

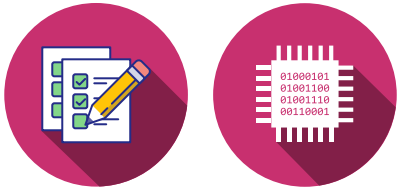
	C		D
0	1	1	0
0	1	0	1
1	0	1	0
1	0	0	1

A

B

$$\overline{C} \oplus \overline{B} \oplus DA$$

Aufgabe 5.1



Darstellung von XOR-Funktionen (benutzten Sie Karnaughtabellen mit 4 Eingängen)

$a \oplus b$

$a \oplus b \oplus c$

Karnaugh Tabellen

Vereinfachung mittels eines Exklusiv-Oder (XOR)

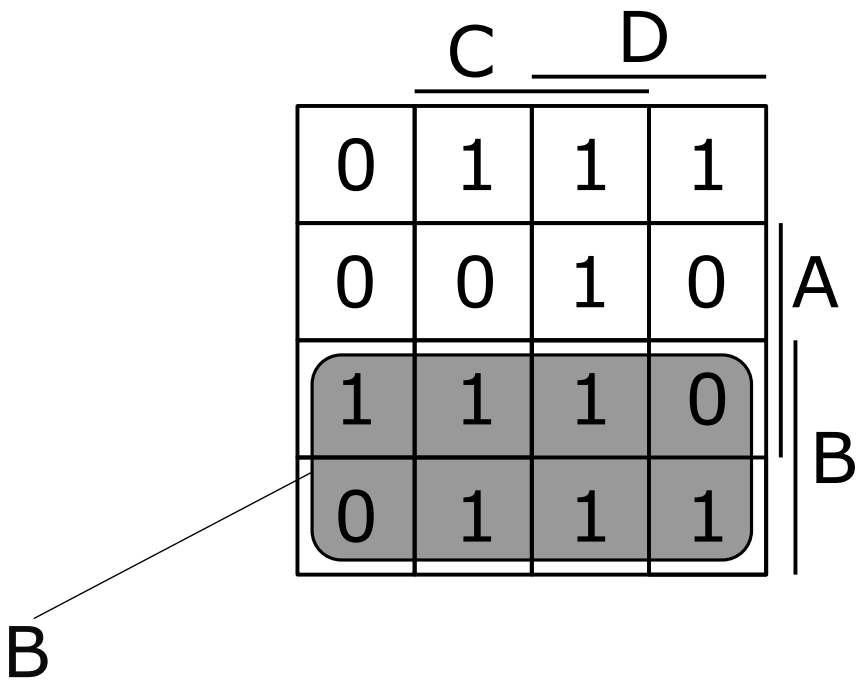
	C	D	
0	1	1	1
0	0	1	0
1	1	1	0
0	1	1	1

A

B

Karnaugh Tabellen

Vereinfachung mittels eines Exklusiv-Oder (XOR)



Karnaugh Tabellen

Vereinfachung mittels eines Exklusiv-Oder (XOR)

	C		D	
0	1	1	1	
0	0	1	0	A
1	1	1	0	
0	1	1	1	B

$$B \oplus \overline{B}D$$

Karnaugh Tabellen

Vereinfachung mittels eines Exklusiv-Oder (XOR)

	C		D	
0	1	1	1	
0	0	1	0	A
1	1	1	0	
0	1	1	1	B

$$B \oplus \overline{B}D \oplus D\overline{C}A$$

Karnaugh Tabellen

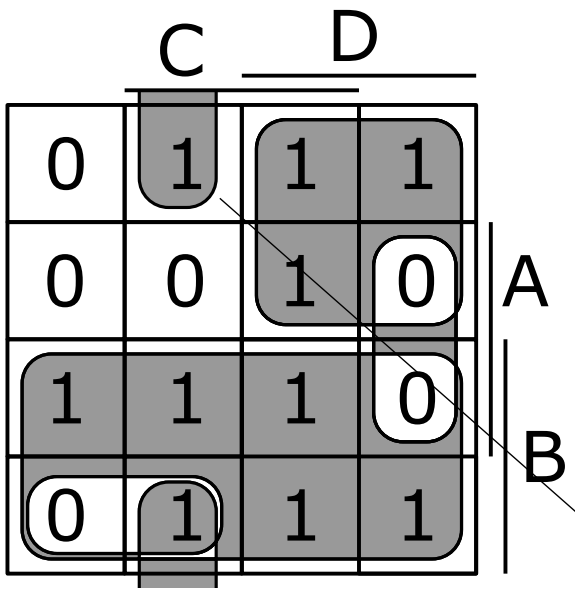
Vereinfachung mittels eines Exklusiv-Oder (XOR)

	C		D	
0	1	1	1	
0	0	1	0	A
1	1	1	0	
0	1	1	1	B

$$B \oplus \overline{B}D \oplus D\overline{C}A \oplus \overline{D}\overline{B}\overline{A}$$

Karnaugh Tabellen

Vereinfachung mittels eines Exklusiv-Oder (XOR)



$$B \oplus \overline{B}D \oplus D\overline{C}A \oplus \overline{D}B\overline{A} \oplus \overline{D}\overline{C}\overline{A}$$

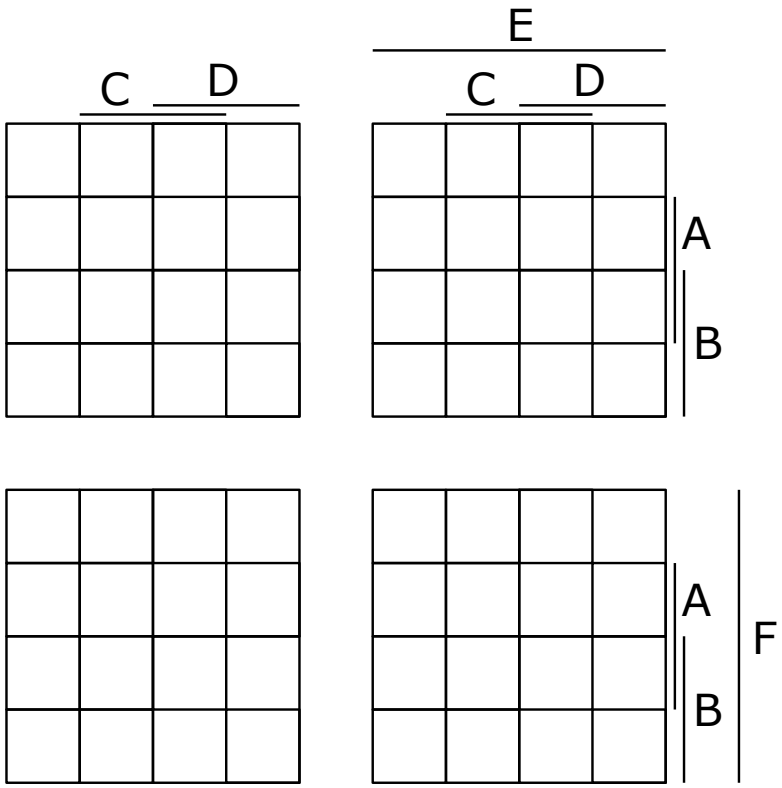
Inhalt

- Karnaugh Tabellen
- Vereinfachung mittels einer Summe von Produkten
- Vereinfachung mittels Exklusiv-ODER Funktionen
- **Funktionen mit einer grossen Anzahl an Eingängen**
  - Algebraische Vereinfachung
  - Vereinfachung in Blöcke von überschaubarer Grösse
  - Iterative Systeme

Karnaugh Tabellen

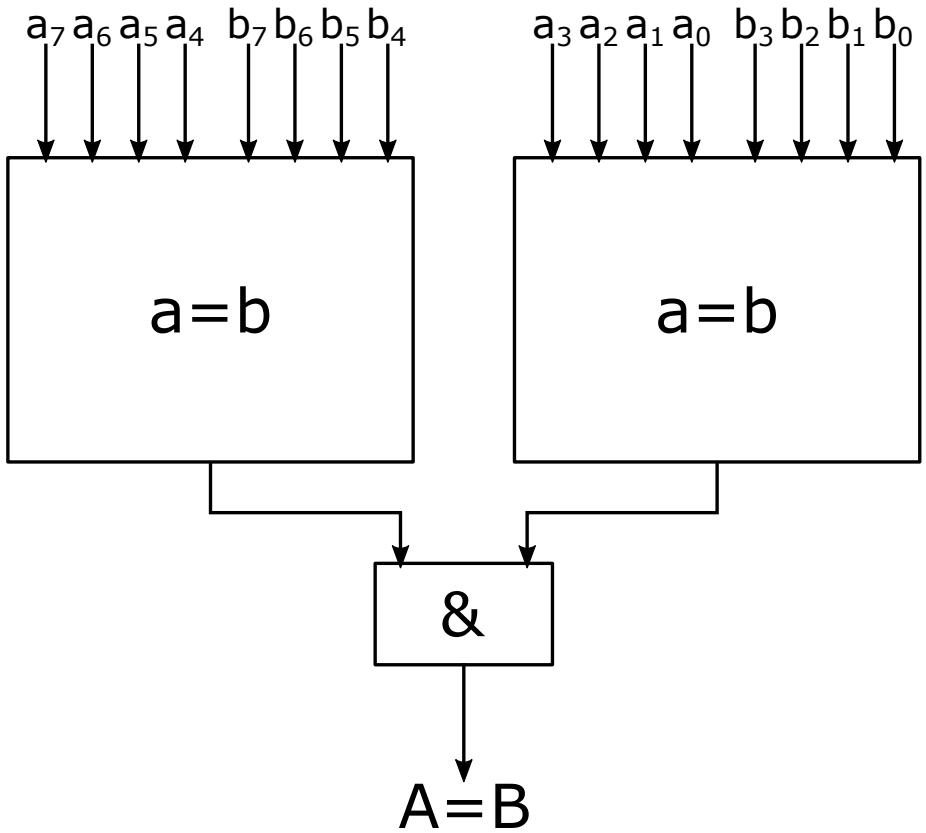
Funktionen mit einer grossen Anzahl an Eingängen

Funktion mit 6 Eingängen



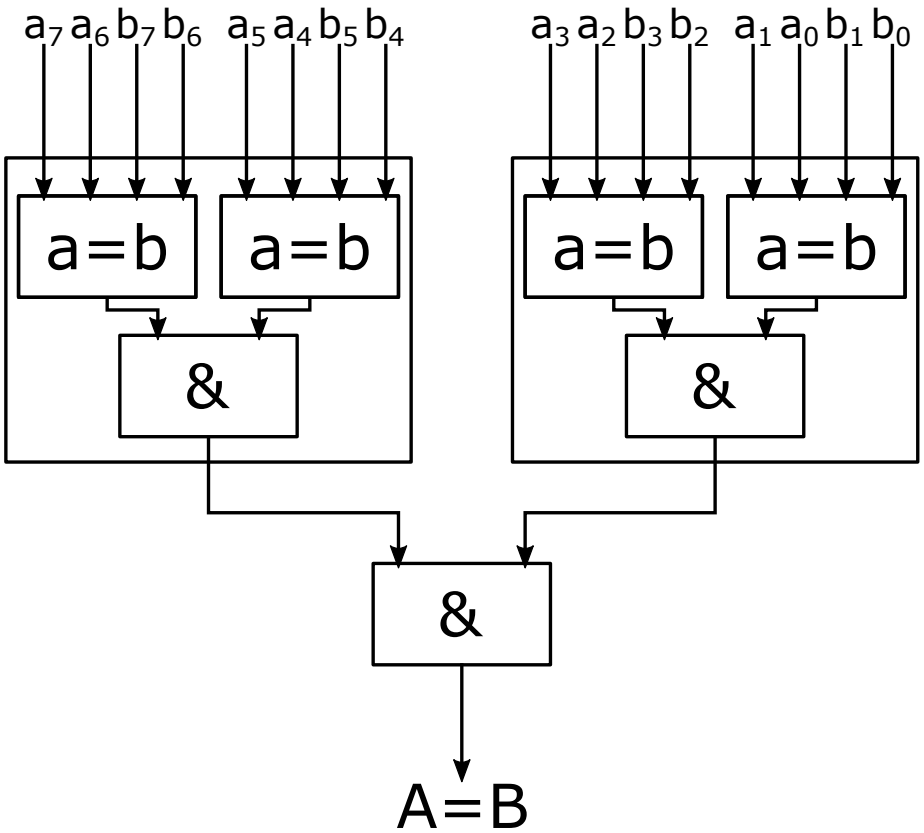
Karnaugh Tabellen

Vereinfachung in Blöcke



Karnaugh Tabellen

Vereinfachung in Blöcke



Karnaugh Tabellen

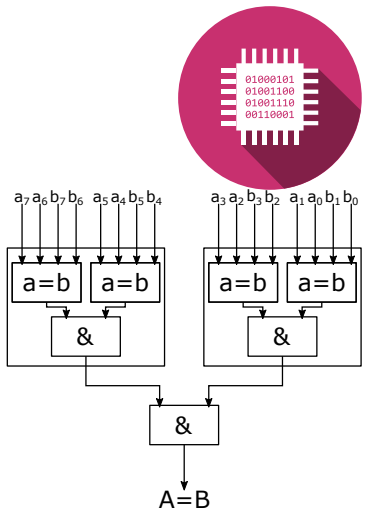
Vereinfachung in Blöcke

A=B, 2x2 Bits

	$\overline{b_0}\,\overline{b_1}$	$\overline{b_0}\,b_1$	$b_0\,b_1$	$b_0\,\overline{b_1}$
$\overline{a_0}\,\overline{a_1}$	1	0	0	0
$\overline{a_0}\,a_1$	0	1	0	0
$a_0\,a_1$	0	0	1	0
$a_0\,\overline{a_1}$	0	0	0	1

a_0

a_1



$$\overline{a_0}\,\overline{b_0}\,\overline{a_1}\,\overline{b_1}+a_0b_0\overline{a_1}\,\overline{b_1}+\overline{a_0}\,\overline{b_0}a_1b_1+a_0b_0a_1b_1$$

Karnaugh Tabellen

Algebraische Vereinfachung

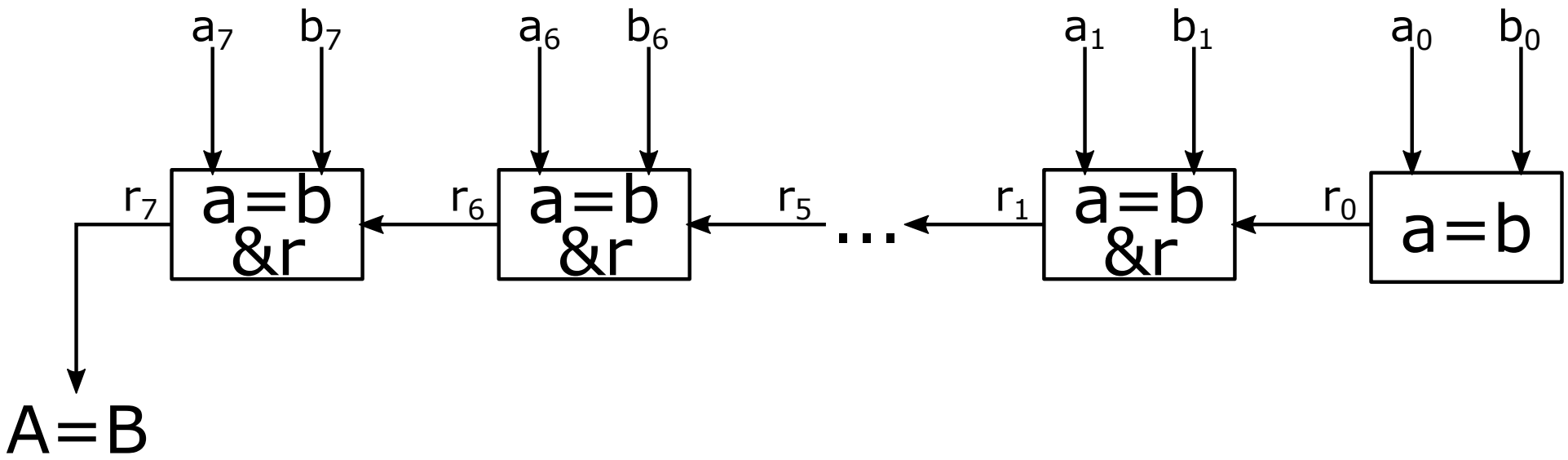
Zahlenvergleich algebraisch

$$A = a_7 * 2^7 + a_6 * 2^6 + \cdots + a_1 * 2^1 + a_0 * 2^0$$
$$B = b_7 * 2^7 + b_6 * 2^6 + \cdots + b_1 * 2^1 + b_0 * 2^0$$

$$A = B \iff (a_7 = b_7) * (a_6 = b_6) * \ldots * (a_1 = b_1) * (a_0 = b_0)$$
$$A = B \iff \overline{(a_7 \oplus b_7)} * \overline{(a_6 \oplus b_6)} * \ldots * \overline{(a_1 \oplus b_1)} * \overline{(a_0 \oplus b_0)}$$
$$A = B \iff \overline{(a_7 b_7 + \overline{a_7}\,\overline{b_7})} * \overline{(a_6 b_6 + \overline{a_6}\,\overline{b_6})} * \ldots * \overline{(a_1 b_1 + \overline{a_1}\,\overline{b_1})} * \overline{(a_0 b_0 + \overline{a_0}\,\overline{b_0})}$$

Karnaugh Tabellen

Iterative Systeme



Karnaugh Tabellen

Iterative Systeme

Iterativer Block

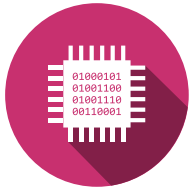
	$\overline{b_i}\,\overline{a_i}$	$\overline{b_i}\,a_i$	$b_i\,a_i$	$b_i\,\overline{a_i}$
r_{i-1}	0	0	0	0
$\overline{r_{i-1}}$	1	0	1	0

$$\overline{a_i}\,\overline{b_i}\,r_{i-1} + a_i b_i r_{i-1}$$

Erster Block unterschiedlich der restlichen: $r_0 = \overline{a_0} * \overline{b_0} + a_0 * b_0$

Erster Block gleich wie die restlichen: $r_{-1} = 1$

Aufgabe 6.1



Zahlenvergleich (A > B)

Referenzen

- [Man78] (Französisch) Gute Presentation
- [Toc92] (Französisch) Vollständige Beispiele
- [Max95] (Englisch) Einfach zum lesen
- [Wak00] (Englisch) Gute Erklärungen Karnaugh + Algorithmen
- [Fle80] (Englisch) Vereinfachungs Algorithmen
- [Kat94] (Englisch) Vereinfachungs Algorithmen

QUESTIONS

CAN BE ASKED BY ANYONE ANYTIME

WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SVCHOST.EXE RUNNING
WHY AREN'T ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY WUBA LUBBA DUB DUB MEANING
WHY IS THERE A WHALE AND A POT FALLING
WHY ARE THERE SO MANY BIRDS IN SWISS
WHY IS THERE SO LITTLE RAIN IN WALLIS
WHY IS WALLIS WEATHER FORECAST ALWAYS WRONG
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REACH UP
HOW FAST IS LIGHTSPEED
WHY ARE OLD KLINGONS DIFFERENT

WHY ARE THERE SQUIRRELS

WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPYDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK

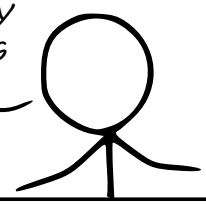
WHY ARE THERE DINOSAUR GHOSTS
WHY AREN'T THERE
WHY IS THERE HELL IF

WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY HAVE DINOSAURS NO FUR
WHY DO IGUANAS DIE

WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE SWISS AFRAID OF DRAGONS

WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON TWITTER
WHY IS HTTPS IMPORTANT

WHY AREN'T MY ARMS GROWING



WHY ARE THERE WEEKS
WHY DO I FEEL DIZZY
WHY IS YKK ON ALL ZIPPERS

WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY IS 42 THE ANSWER TO EVERYTHING
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER
WHY IS MARVIN ALWAYS SO SAD
WHY IS THERE A SWARM OF ANTS
WHY IS THERE PILEGRIM
WHY IS THERE LAVA
WHY ARE THERE SO MANY CROWS IN ROCHESTER
WHY IS TO BE OR NOT TO BE FUNNY
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY ARE THERE ANTS IN MY LAPTOP
WHY ARE DOGS AFRAID OF FIR
WHY IS THERE NO KING IN EN

WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE FEMALE

WHY ARE THERE GHOSTS

WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE FPGA's EVERYWHERE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOCKS
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO PEOPLE DIE
WHY IS LIFE SO

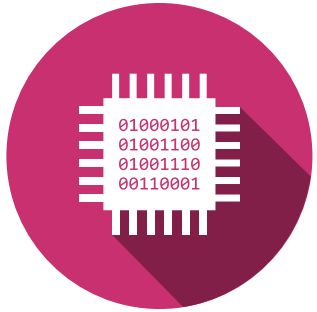
WHAT IS https://xkcd.com/1256/

WHY DO THEY SAY T-MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET

WHY AREN'T THERE GUNS IN

Hes·so VALAIS WALLIS

Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften



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